Space complexity: L, PSPACE, and NL

Exercise sheet 11

Exercise 11.1. Let A, B, and C be languages. Prove the following statements about logspace reductions:

(a) If $A \leq_L B$ and $B \leq_L C$, then $A \leq_L C$.

(b) If $A \leq_L B$ and $B \in \mathrm{L}$, then $A \in \mathrm{L}$.

Exercise 11.2. Prove that if $A, B \in \mathrm{L}$, then $A \cup B \in \mathrm{L}$ and $A \cap B \in \mathrm{L}$.

Exercise 11.3. Consider the alphabet $\{(,)\}$ of *parentheses*. Prove that the language of all *balanced-parentheses* words is in L.

A *nondeterministic* Turing machine may have several possible transitions for a state and observed symbol. To describe all its possible behaviours, imagine following every legal choice: whenever a configuration has several next moves, the computation branches, with one branch for each move. A path through this tree is a computation. The machine accepts an input if at least one computation halts in the accepting state, and rejects if all computations reject. It runs in polynomial time if every computation halts within a polynomial number of steps.

Exercise 11.4. * Prove that $L \in \mathrm{NP}$ if and only if a polynomial-time nondeterministic Turing machine decides L.

Exercise 11.5. * Let NEXP be the class of languages decided by exponential-time nondeterministic Turing machines. Prove that if $\mathrm{P} = \mathrm{NP}$, then $\mathrm{EXP} = \mathrm{NEXP}$.

Exercise 11.6. Prove that $\mathrm{NL} \subseteq \mathrm{P}$.

Exercise 11.7. Prove that DIRECTED-REACHABILITY is in NL. Here the input is a directed graph G and two vertices s and t, and the question is whether there is a directed path from s to t. The lecture also states that this problem is NL-complete under logspace reductions. Explain why a deterministic logspace algorithm for it would imply $\mathrm{L} = \mathrm{NL}$, and whether we know that this equality holds.

Exercise 11.8. * Prove that NON-BIPARTITE, the problem of deciding whether an undirected graph is not bipartite, is in NL. Does this also prove that BIPARTITE is in NL? Briefly explain.

Exercise 11.9. * Prove that $\mathrm{P} \neq \mathrm{SPACE}(n)$.